

Alexander Lopez

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EDUCATION

Northeastern University, Khoury College of Computer Science

September 2024 - May 2028

- Combined BSE in Computer Science and Mathematics with a Concentration in AI

Cumulative GPA: 3.74/4.0

- **Programming Languages:** C++ | C | Python | Java | JavaScript | OCaml

Major GPA: 3.94/4.0

- **Frameworks and Tools:** PyTorch | React | SQL | AWS | Pandas | HTML/CSS

- **Relevant Courses:** Operating Systems (C, Built an OS from scratch) | Data Structures and Algorithms (Java) | Advanced Probability and Statistics | Number Theory I | Group Theory | Numerical Methods For Differential Equations

Hopkins School

June 2024

- **Extracurriculars:** Math Team | Science Olympiad | Model UN | Diversity Board | Varsity Wrestling (Captain), Football, and Track

EXPERIENCE

Expandya

June 2025 - September 2025

SWE Intern

Westport, CT

- Contributed to the development of database architectures using Python and C++ to efficiently query and manage datasets exceeding (50M+ rows) by implementing optimized indexing and query caching to reduce system latency and improve retrieval speeds by over 1,000% for high-concurrency client environments
- Spearheaded weekly technical presentations for clientele, delivering actionable data-driven insights and possible strategies to drive key initiatives for customer acquisition and retention

O'Hern Lab at Yale University

June 2023 – September 2023

Research Assistant

New Haven, CT

- Optimized backend scripts to analyze and process structural data for cancer-causing protein variants (10,000+) utilizing Yale's high-performance computing (HPC) clusters to generate and run protein simulations based on models produced by the neural networks AlphaFold and NetSurfP to determine the pathogenic impact based on their protein structure and folding
- Conducted weekly meetings using PowerPoint presentations to share progress updates, highlight key findings, and collaborate on finding new areas of research and ideas, culminating in a presentation to the department of Computational Biology at Yale

Northeastern University

September 2025 - Present

Teaching Assistant

Boston, MA

- Teach weekly recitation classes for a Computer Systems course covering low-level circuits, logic gates, and programming languages such as Assembly and C; assist students in debugging homework and projects during weekly office hours

Pi Kappa Alpha Fraternity Kappa Delta

January 2026 – Present

Brotherhood Chair

Boston, MA

- Collaborate with other board members to develop new leadership initiatives that promote mentorship, networking, and expanded professional development opportunities for the growth of fellow brothers

PROJECTS

IMC Prosperity 3

March 2025 - April 2025

- Finished top 2% (184 out of 12,000) in a global algorithmic trading competition; working alongside my teammates, researched and designed various trading opportunities that involved market making, scalping, locational arbitrage, and mean-reversion of alpha to optimize both execution and risk-adjusted returns

Momentum Trading Bot

July 2024 – August 2024

- Leveraged Alpaca's API to code a day trading bot that calculates the Relative Strength Index of the Dow Jones 30 and invests in the stock with the lowest RSI if it is less than 20; achieved an initial CAGR of 8.29% and a highest of 20.37%
- Learned about different trading strategies (momentum, mean reversion, and news analysis), order types (bracket, limit, and market), and position management strategies (various stop-loss and take-profit thresholds)

QHO Bot

September 2023 – October 2023

- Developed a Python script using the Selenium library for quick queue registration on the office hours website for my high school's courses, to give students an early position in the queue and quickly get help from their teacher during office hours

SKILLS AND INTERESTS

Skills: Microsoft Office Suite | Analytical | Communication | Project Management | Problem-Solving | Adaptable | Attention to Detail

Interests: Running (training for my second marathon) | Indoor and Outdoor Rock Climbing | Manchester United (and all things soccer)